

```
import pandas as pd
df = pd.read_csv("/content/train.csv")
print(df)
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	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	
889	Behr, Mr. Karl Howell	male	26.0	0	
890	Dooley, Mr. Patrick	male	32.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S
..	...	...	...	...	...
886	0	211536	13.0000	NaN	S
887	0	112053	30.0000	B42	S
888	2	W./C. 6607	23.4500	NaN	S
889	0	111369	30.0000	C148	C
890	0	370376	7.7500	NaN	Q

[891 rows x 12 columns]

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import pandas as pd

# 1. Display first 10 rows
print(df.head(10))

# 2. Display last 7 rows
print(df.tail(7))

# 3. Show 5 random rows
print(df.sample(5))

# 4. Total number of rows and columns
print(df.shape)

# 5. Rows and Columns separately
print("Total Columns:", df.shape[1])

# 6. Print all column names
print(df.columns)

# 7. Display complete information
df.info()

# 8. Statistical summary
print(df.describe())

# 9. Create df_small with first 20 rows
df_small = df.head(20)
print(df_small)

# 10. Create DataFrame with Name, Age, Sex, Fare
df_new = df[['Name', 'Age', 'Sex', 'Fare']]
print(df_new)
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# section-B - Selecting data
# 11. Select only the Age column
print(df['Age'])

# 12. Select Name, Pclass, and Survived columns
print(df[['Name', 'Pclass', 'Survived']])

# 13. Select first row using iloc
print(df.iloc[0])

# 14. Select rows 10 to 20 using iloc
print(df.iloc[10:21])

# 15. Select rows 50 to 60 and columns Name and Fare
print(df.loc[50:60, ['Name', 'Fare']])

# 16. Select passenger name in the 100th row
print(df.loc[99, 'Name'])

# 17. Select last 15 rows
print(df.tail(15))

# 18. Select passengers whose names start with A
print(df[df['Name'].str.startswith('A', na=False)])

# 19. Fare values of first 25 passengers
print(df.loc[:24, 'Fare'])

# 20. Rows where PassengerId is between 200 and 220
print(df[(df['PassengerId'] >= 200) & (df['PassengerId'] <= 220)])

# section-C filtering data
# 21. Passengers whose age is greater than 30
print(df[df['Age'] > 30])

# 22. All female passengers
print(df[df['Sex'] == 'female'])

# 23. Male passengers traveling in first class
print(df[(df['Sex'] == 'male') & (df['Pclass'] == 1)])

# 24. Passengers whose fare is greater than 100
print(df[df['Fare'] > 100])

# 25. Passengers whose age is less than 18
print(df[df['Age'] < 18])

# 26. Passengers who embarked from S
print(df[df['Embarked'] == 'S'])

# 27. Passengers who survived
print(df[df['Survived'] == 1])

# 28. Female passengers who survived
print(df[(df['Sex'] == 'female') & (df['Survived'] == 1)])

# 29. Passengers whose age is between 20 and 40
print(df[(df['Age'] >= 20) & (df['Age'] <= 40)])

# 30. Passengers whose age is missing
print(df[df['Age'].isnull()])

# section-D - Adding new column
# 31. Create Family_Size column
df['Family_Size'] = df['SibSp'] + df['Parch'] + 1
print(df[['Name', 'Family_Size']])

# 32. Create Fare_with_Tax column
df['Fare_with_Tax'] = df['Fare'] * 1.18
print(df[['Name', 'Fare', 'Fare_with_Tax']])

# 33. Create Is_Child column
df['Is_Child'] = df['Age'] < 18
print(df[['Name', 'Age', 'Is_Child']])
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# 34. Create Name_Length column
df['Name_Length'] = df['Name'].str.len()
print(df[['Name', 'Name_Length']])

# 35. Create Fare_Category column
def fare_category(fare):
    if fare > 100:
        return "Very High"
    elif fare > 50:
        return "High"
    else:
        return "Low"

df['Fare_Category'] = df['Fare'].apply(fare_category)

print(df[['Name', 'Fare', 'Fare_Category']])

# section-E updating value
# 36. Increase all fares by 10%
df['Fare'] = df['Fare'] * 1.10
print(df[['Name', 'Fare']])

# 37. Replace missing values in Embarked with 'Unknown'
df['Embarked'] = df['Embarked'].fillna('Unknown')
print(df['Embarked'])

# 38. Convert Sex values from male/female to M/F
df['Sex'] = df['Sex'].replace({
    'male': 'M',
    'female': 'F'
})
print(df['Sex'])

# 39. Set fare of first passenger to 999
df.loc[0, 'Fare'] = 999
print(df.loc[0])

# 40. Increase age of all passengers by 1 year
df['Age'] = df['Age'] + 1
print(df[['Name', 'Age']])

# section F deleting data
# 41. Delete Cabin column
df = df.drop('Cabin', axis=1)
print(df.columns)

# 42. Delete rows where Age is missing
df = df.dropna(subset=['Age'])
print(df)

# 43. Delete first 5 rows
df = df.iloc[5:]
print(df)

# 44. Delete duplicate rows
df = df.drop_duplicates()
print(df)

# section G handling missing values
# 45. Count missing values in each column
print(df.isnull().sum())

# 46. Total missing values in entire dataset
print(df.isnull().sum().sum())

# 47. Fill missing Age values with mean age
df['Age'] = df['Age'].fillna(df['Age'].mean())
print(df['Age'])

# 48. Fill missing Embarked values with mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
print(df['Embarked'])
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# 49. Drop all rows containing at least one missing value
df_clean = df.dropna()
print(df_clean)

# 50. Compare shape before and after handling missing values
print("Original Shape:", df.shape)
print("After Cleaning:", df_clean.shape)
```

802	803	1	1
856	857	1	1

	Name	Sex	Age	SibSp	\
27	Fortune, Mr. Charles Alexander	male	19.00	3	
31	Spencer, Mrs. William Augustus (Marie Eugenie)	female	NaN	1	
88	Fortune, Miss. Mabel Helen	female	23.00	3	
118	Baxter, Mr. Quigg Edmond	male	24.00	0	
195	Lurette, Miss. Elise	female	58.00	0	
215	Newell, Miss. Madeleine	female	31.00	1	
258	Ward, Miss. Anna	female	35.00	0	
268	Graham, Mrs. William Thompson (Edith Junkins)	female	58.00	0	
269	Bissette, Miss. Amelia	female	35.00	0	
297	Allison, Miss. Helen Loraine	female	2.00	1	
299	Baxter, Mrs. James (Helene DeLaudeniére Chaput)	female	50.00	0	
305	Allison, Master. Hudson Trevor	male	0.92	1	
306	Fleming, Miss. Margaret	female	NaN	0	
307	Penasco y Castellana, Mrs. Victor de Satode (M...	female	17.00	1	
311	Ryerson, Miss. Emily Borie	female	18.00	2	
318	Wick, Miss. Mary Natalie	female	31.00	0	
319	Spedden, Mrs. Frederic Oakley (Margaretta Corn...	female	40.00	1	
325	Young, Miss. Marie Grace	female	36.00	0	
332	Graham, Mr. George Edward	male	38.00	0	
334	Frauenthal, Mrs. Henry William (Clara Heinshei...	female	NaN	1	
337	Burns, Miss. Elizabeth Margaret	female	41.00	0	
341	Fortune, Miss. Alice Elizabeth	female	24.00	3	
373	Ringhini, Mr. Sante	male	22.00	0	
377	Widener, Mr. Harry Elkins	male	27.00	0	
380	Bidois, Miss. Rosalie	female	42.00	0	
390	Carter, Mr. William Ernest	male	36.00	1	
393	Newell, Miss. Marjorie	female	23.00	1	
435	Carter, Miss. Lucile Polk	female	14.00	1	
438	Fortune, Mr. Mark	male	64.00	1	
498	Allison, Mrs. Hudson J C (Bessie Waldo Daniels)	female	25.00	1	
505	Penasco y Castellana, Mr. Victor de Satode	male	18.00	1	
527	Farthing, Mr. John	male	NaN	0	
537	LeRoy, Miss. Bertha	female	30.00	0	
544	Douglas, Mr. Walter Donald	male	50.00	1	
550	Thayer, Mr. John Borland Jr	male	17.00	0	
557	Robbins, Mr. Victor	male	NaN	0	
581	Thayer, Mrs. John Borland (Marian Longstreth M...	female	39.00	1	
609	Shutes, Miss. Elizabeth W	female	40.00	0	
659	Newell, Mr. Arthur Webster	male	58.00	0	
660	Frauenthal, Dr. Henry William	male	50.00	2	
679	Cardeza, Mr. Thomas Drake Martinez	male	36.00	0	
689	Madill, Miss. Georgette Alexandra	female	15.00	0	
698	Thayer, Mr. John Borland	male	49.00	1	
700	Astor, Mrs. John Jacob (Madeleine Talmadge Force)	female	18.00	1	
708	Cleaver, Miss. Alice	female	22.00	0	
716	Endres, Miss. Caroline Louise	female	38.00	0	
730	Allen, Miss. Elisabeth Walton	female	29.00	0	
737	Lesurer, Mr. Gustave J	male	35.00	0	
742	Ryerson, Miss. Susan Parker "Suzette"	female	21.00	2	
763	Carter, Mrs. William Ernest (Lucile Polk)	female	36.00	1	
779	Robert, Mrs. Edward Scott (Elisabeth Walton Mc...	female	43.00	0	
802	Carter, Master. William Thornton II	male	11.00	1	
856	Wick, Mrs. George Dennick (Mary Hitchcock)	female	45.00	1	

